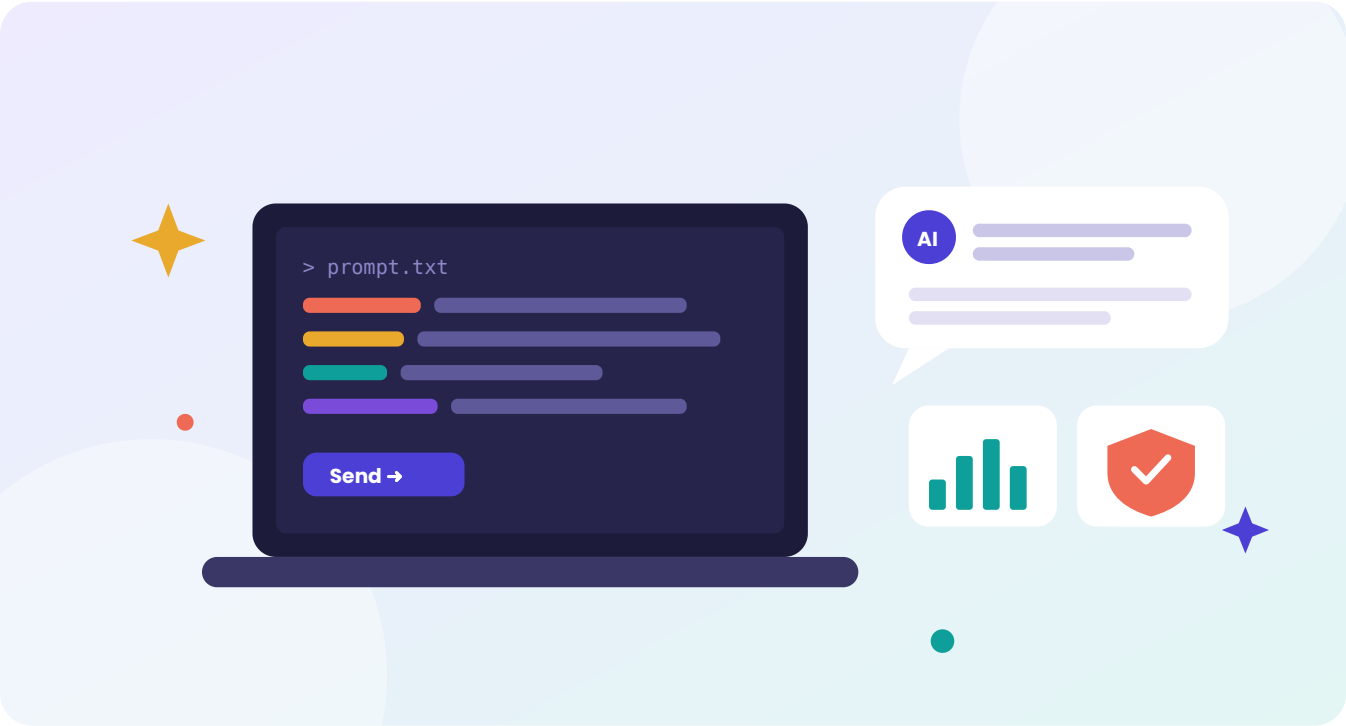
STUDENT CAREER GUIDE · 2026 EDITION

Prompt Engineering for Your IT Career

A friendly, practical handbook for students stepping into tech — learn how to talk to AI tools clearly so they help you code, test, secure, analyse and design better.

Python | Cybersecurity | Selenium & Playwright | Full Stack | Data Analytics | Data Science & ML | UI/UX



What's inside

Read it front to back once, then keep the career sections handy as a prompt cookbook.

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01

Introduction

Talking to AI is a skill — and you can learn it.



Tools like ChatGPT, Claude, Gemini and GitHub Copilot are now part of everyday work in IT. You type something, the AI replies. Simple — but the **quality of the reply depends heavily on the quality of what you type**. That typed instruction is called a **prompt**.

Prompt engineering is the skill of writing prompts that are clear, specific and well-structured, so the AI gives you useful, accurate, ready-to-use output on the first or second try — instead of the tenth.

A prompt is…
The instruction, question or context you give an AI model: a task, some background, maybe examples.

An LLM is…
A Large Language Model — an AI trained on huge amounts of text that predicts useful replies to your prompt.

Engineering means…
Designing, testing and improving your prompts — just like you would design and debug code.

A simple way to think about it

Imagine you have a very smart new intern who knows a lot but knows *nothing* about your project. If you say "make a login page", you'll get something generic. If you explain the tech stack, the users, the design style and what "done" looks like, you'll get something close to what you need. AI works the same way.

💡 **Remember** AI is a **co-pilot, not an autopilot**. You are still responsible for understanding, checking and testing everything it produces. Recruiters hire people who can think — AI just helps you think faster.

Who this guide is for

- Students and freshers starting an IT course or looking for their first tech job
- Career switchers moving into Python, testing, full stack, data or design roles
- Anyone who uses AI tools but feels the answers are "okay, not great"

02

Why Prompt Engineering?

The same AI tool gives very different results to different people.



⚡ Save hours every week

Get working code, test cases, SQL queries and documentation drafts in minutes instead of hours.

🎯 Get accurate answers

Clear prompts reduce wrong, generic or "hallucinated" answers — so less time fixing AI mistakes.

📚 Learn faster

Use AI as a 24×7 tutor that explains concepts at your level, quizzes you and reviews your code.

💼 Stand out in interviews

Companies now ask "How do you use AI in your work?" A thoughtful answer shows maturity.

🧩 Works in every IT role

Developers, testers, analysts, security engineers and designers all use AI daily.

🚀 Future-proof skill

Tools change, but the habit of giving clear context and checking results stays valuable.

The difference, in one example

× WEAK PROMPT

```
write code for student marks
```

✓ STRONG PROMPT

```
Act as a Python mentor for a beginner.
Write a Python program that reads student names and marks for 3 subjects
from a CSV file, calculates the total and average, and prints the top 3 students.
Use only the built-in csv module, add comments on each step, and show
a sample CSV with 5 rows so I can test it.
```

The second prompt tells the AI **who to be**, **what to build**, **what tools to use**, **what level to explain at** and **how to show the output**. That's prompt engineering.

03

Basics of Prompt Engineering

One formula, a few techniques, lots of practice.

The 6-part prompt formula

You don't need all six every time, but the more important the task, the more parts you should include.

Role + **Context** + **Task** + **Constraints** + **Format** + **Examples**

Part	What it means	Example phrase
Role	Who the AI should act as	"Act as a senior QA engineer..."
Context	Background: project, audience, your level	"I'm a fresher building a college project in Django..."
Task	The exact thing you want done	"Write 10 test cases for the login page..."
Constraints	Rules, limits, tools, versions	"Use Python 3.12, no external libraries, under 50 lines."
Format	How the answer should look	"Give the output as a table with columns ID, Steps, Expected Result."
Examples	A sample of what good looks like	"Here is one test case in my format: ..."

Core techniques every student should know

1 · Zero-shot prompting

Ask directly without examples. Good for simple, common tasks.

```
Explain what a REST API is in 5 lines.
```

2 · Few-shot prompting

Give 1–3 examples so the AI copies your pattern, style or format exactly.

3 · Step-by-step reasoning

Ask the AI to work through the problem step by step before answering. Great for logic, maths and debugging.

4 · Role prompting

"Act as an interviewer / reviewer / tutor." The role sets the tone, depth and vocabulary.

5 · Iterative refinement

Treat the first answer as a draft. Follow up: "make it shorter", "add error handling", "explain line 12".

6 · Ask the AI to ask you

"Before you start, ask me any questions you need." This fills gaps you didn't know you had.

Few-shot example

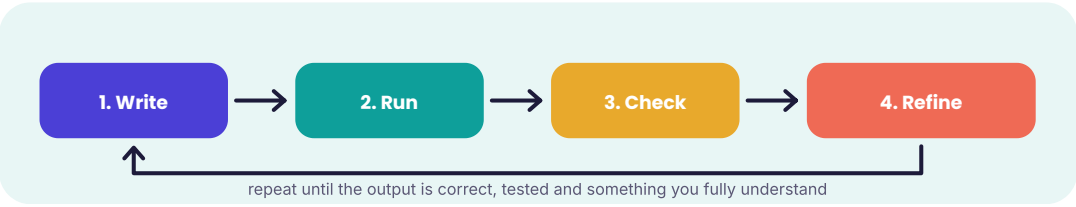
PROMPT

```
Convert these features into user stories in the same format.

Example:
Feature: password reset
Story: As a registered user, I want to reset my password by email,
so that I can access my account if I forget it.

Now do the same for: (1) profile photo upload (2) order history (3) dark mode
```

The prompt improvement loop

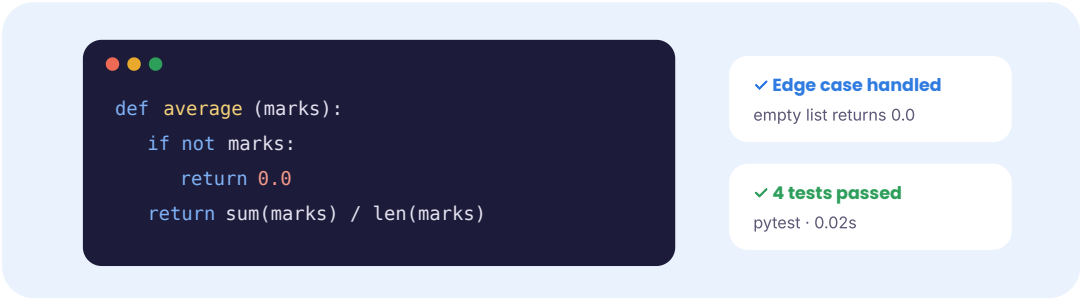


⚠ **Never paste** passwords, API keys, company source code, customer data or personal information into public AI tools. Mask them (e.g. `API_KEY=xxxx`) or use your organisation's approved tools.

04

Prompt Engineering for Python

Your always-available coding mentor and pair programmer.



Where AI helps Python learners

Learn concepts

Loops, OOP, decorators, generators explained with simple examples.

Debug errors

Paste the traceback, get the cause and the fix explained.

Write & refactor

Generate functions, clean messy code, add type hints and docstrings.

PROMPT · UNDERSTAND A CONCEPT

```
I'm a Python beginner. Explain list comprehensions using a real-life analogy,
then show 3 examples from easy to medium. After each example, give me
one small exercise to try myself. Don't show the answers until I ask.
```

PROMPT · DEBUG AN ERROR

```
My Python 3.12 script throws this error:
KeyError: 'email'  (line 18)
Here is the code: [paste code]
Explain in simple words why this happens, show the fixed code,
and suggest how to avoid this error in future (e.g. dict.get()).
```

PROMPT · CODE REVIEW

```
Act as a senior Python developer reviewing a junior's code.
Review the code below for readability, PEP 8 style, edge cases and performance.
Give feedback as a numbered list (most important first), then show an improved version.
[paste code]
```

PROMPT · WRITE TESTS

```
Write pytest unit tests for the function below. Cover normal cases,
edge cases (empty input, negative numbers, wrong types) and use
@pytest.mark.parametrize where it makes sense.
[paste function]
```

💡 **Pro tip** Always mention your **Python version** and **libraries** (e.g. pandas 2.x, Django 5). Then **run the code yourself** — never submit AI code you haven't executed and understood.

05

Prompt Engineering for Cybersecurity

Learn faster, defend better — always ethically.



⚖ Ethics first Only test systems you **own** or have **written permission** to test (labs like TryHackMe, HackTheBox, DVWA, your own VMs). AI tools will — and should — refuse requests to attack real systems or build malware. Focus your prompts on **learning, defence and analysis**.

Where AI helps security learners

Concepts & certifications

OWASP Top 10, networking, cryptography basics, Security+ / CEH style revision.

Secure code review

Spot SQL injection, XSS and weak authentication in your own code and learn the fix.

Log & alert analysis

Understand suspicious log entries, write detection rules and incident summaries.

Hardening & policies

Checklists for securing Linux servers, cloud accounts and web apps.

PROMPT · LEARN A VULNERABILITY

```
Act as a cybersecurity instructor. Explain SQL injection to a fresher:
what it is, why it happens, a harmless example of vulnerable Python code,
and the secure version using parameterised queries. End with a 5-question quiz.
```

PROMPT · SECURE CODE REVIEW

```
Review this Flask login route from my college project for security issues
(OWASP Top 10). For each issue give: severity (High/Medium/Low), why it's risky,
and the fixed code. Present it as a table. [paste code]
```

PROMPT · LOG ANALYSIS

```
I'm a SOC analyst trainee. Here are 20 lines of Linux auth.log (IPs masked).
Identify any signs of brute-force or suspicious activity, explain what each
pattern means, and suggest next steps for investigation. [paste logs]
```

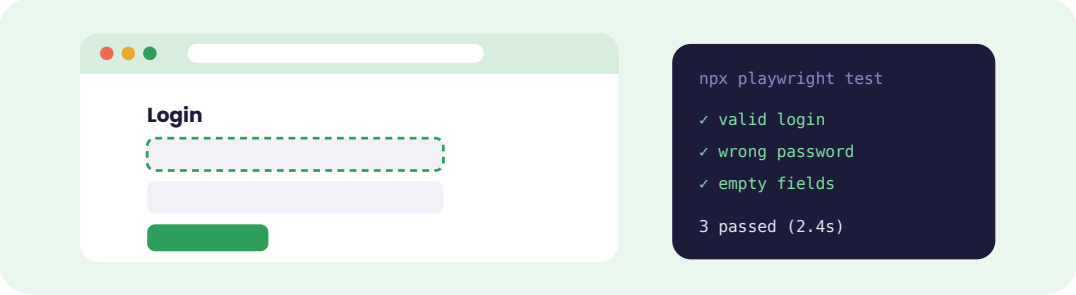
PROMPT · HARDENING CHECKLIST

```
Create a step-by-step hardening checklist for a fresh Ubuntu 24.04 web server
running Nginx. Group by: users & SSH, firewall, updates, logging, web server.
Include the command for each step and explain why it matters.
```

06

Software Testing — Selenium & Playwright

From manual test cases to stable automation scripts.



Where AI helps testers

Test design

Test scenarios, boundary values, negative cases, test data.

Automation

Selenium / Playwright scripts, Page Object Model, assertions.

Debugging

Locator issues, timeouts, flaky tests, CI failures.

PROMPT · MANUAL TEST CASES

```
Act as a QA engineer. Write test cases for an e-commerce login page
(email + password, "Remember me", "Forgot password").
Include positive, negative and boundary cases.
Format as a table: TC ID | Scenario | Steps | Test Data | Expected Result | Priority.
```

PROMPT · SELENIUM SCRIPT

```
Write a Selenium 4 test in Java with TestNG for the login flow below.
Use the Page Object Model, explicit waits (WebDriverWait) instead of Thread.sleep,
and assert that the dashboard heading is visible after login.
Page details: email field id="email", password id="pwd", button text "Sign in".
```

PROMPT · PLAYWRIGHT SCRIPT

```
Write a Playwright test in TypeScript for the same login flow.
Use getByRole / getByLabel locators, the built-in auto-waiting and expect().
Add one test for invalid password that checks the error message.
Also show the playwright.config.ts settings to run in Chromium and Firefox.
```

PROMPT · FIX A FLAKY TEST

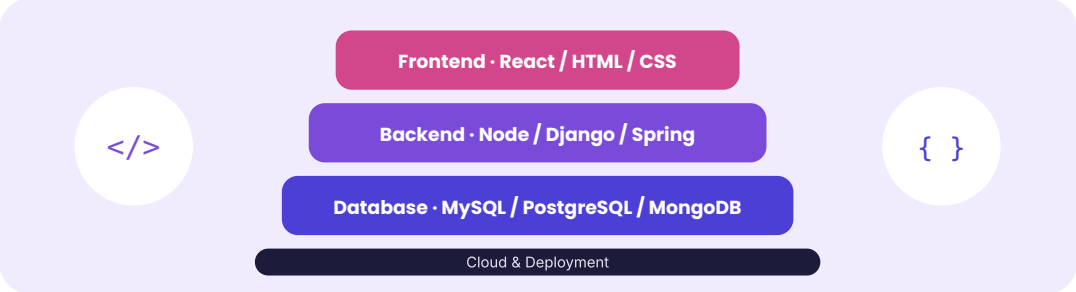
```
This Playwright test passes locally but fails ~30% of the time in GitHub Actions
with "Timeout 30000ms exceeded waiting for locator". Here is the test and the error.
List the most likely causes in order and show the corrected test. [paste code + error]
```

💡 **Always tell the AI** the **tool + version** (Selenium 4, Playwright 1.x), your **language + framework** (Java/TestNG, Python/pytest, TypeScript), the **real HTML** of the element for stable locators, and your **design pattern** (POM, fixtures).

07

Prompt Engineering for Full Stack Development

Frontend, backend, database and deployment — one prompt at a time.



Start with the big picture

PROMPT · PLAN A PROJECT

```
I'm building a "College Event Booking" app as my portfolio project
using React + Node.js (Express) + MongoDB. I'm a fresher.
Suggest: (1) main features for an MVP, (2) folder structure,
(3) database collections with fields, (4) list of REST API endpoints.
Keep it simple enough to finish in 3 weeks.
```

Then build piece by piece

PROMPT · BACKEND API

```
Write an Express.js route for POST /api/bookings that validates
eventId and userId, checks seat availability in MongoDB (Mongoose),
returns proper HTTP status codes (201, 400, 404, 409) and uses async/await
with try/catch. Add comments explaining each step.
```

PROMPT · FRONTEND COMPONENT

```
Create a responsive React component (functional, hooks) called EventCard
that shows event title, date, venue, seats left and a "Book now" button.
Disable the button when seats left = 0. Style with Tailwind CSS.
Also show how to render a list of EventCards from an API call.
```

PROMPT · EXPLAIN UNFAMILIAR CODE

```
Explain this code line by line like I'm new to React.
Then tell me what would break if I removed the dependency array in useEffect.
[paste code]
```

✓ Good habits

- Ask for one feature at a time
- Share relevant files, not your whole project
- Ask "what are the security risks in this code?"

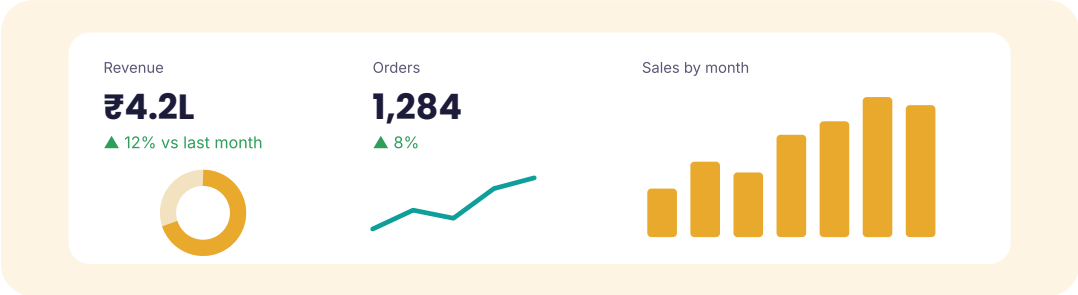
× Avoid

- Copying a whole AI-generated app you can't explain
- Mixing stacks in one prompt ("React or Angular, whatever")
- Committing secrets like `.env` keys

08

Prompt Engineering for Data Analytics

Turn raw data into clear answers with SQL, Excel and BI tools.



Give the AI your table structure

AI can't see your database. The single biggest improvement for data prompts is to **describe your tables and columns** (or paste a few sample rows with fake data).

PROMPT · SQL QUERY

```
I use MySQL 8. Tables:
orders(order_id, customer_id, order_date, amount, status)
customers(customer_id, name, city, signup_date)
Write a query to find the top 5 cities by total revenue in 2025,
counting only orders with status = 'delivered'. Explain each clause.
```

PROMPT · EXCEL FORMULA

```
In Excel, column A has dates, B has product names, C has sales amounts.
Give me a formula to total sales of "Laptop" in March 2026 only.
Show both a SUMIFS version and explain how it works.
```

PROMPT · POWER BI / DAX

```
In Power BI, I have a Sales table with Date, Region, Revenue.
Write a DAX measure for Month-over-Month revenue growth %.
Explain it simply and tell me which visual best shows it.
```

PROMPT · INSIGHTS & STORYTELLING

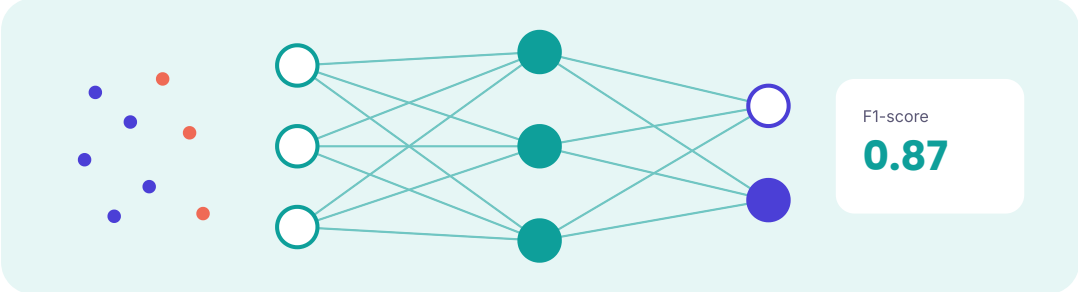
```
Here is a summary table of monthly sales by region [paste].
Act as a business analyst. Give me 5 key insights, 2 possible reasons
for the drop in the South region, and 3 recommendations —
written for a non-technical manager in under 150 words.
```

💡 **Pro tip** Always **verify the numbers yourself**. AI can write the query, but run it on your real data and sanity-check totals. Never paste real customer data — use anonymised or sample rows.

09

Prompt Engineering for Data Science & ML

From messy dataset to trained, evaluated model.



The ML workflow, with AI at each step

Stage	Example ask
Understand problem	"Is this a classification or regression problem? Which metric should I use and why?"
EDA	"Write pandas + seaborn code to explore missing values, distributions and correlations."
Cleaning & features	"Suggest feature engineering ideas for this churn dataset and explain each."
Modelling	"Build a scikit-learn Pipeline comparing Logistic Regression and Random Forest."
Evaluation	"Explain this confusion matrix and whether my model is overfitting."
Communicate	"Summarise these results for a non-technical stakeholder in 5 bullet points."

PROMPT · EDA PLAN

```
I have a CSV of 10,000 telecom customers with columns:
tenure, monthly_charges, contract_type, payment_method, internet_service, churn (Yes/No).
Act as a data science mentor. Give me an EDA plan and the Python code
(pandas, matplotlib/seaborn) for each step, with a one-line note on what to look for.
```

PROMPT · MODEL BUILDING

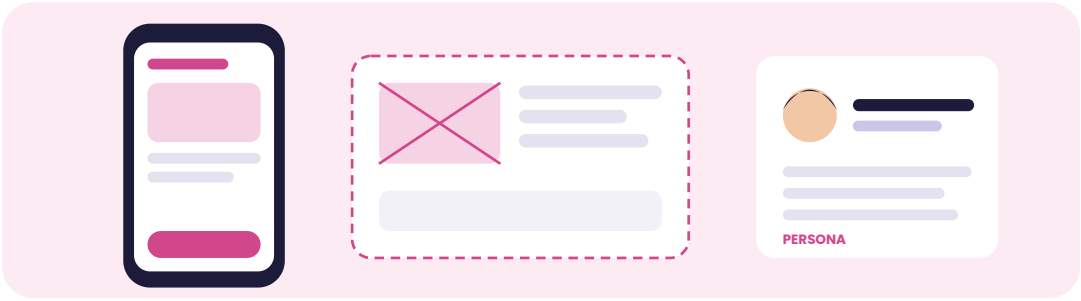
```
Using scikit-learn, write a Pipeline that: imputes missing values,
one-hot encodes categorical columns, scales numeric columns, and trains a
RandomForestClassifier. Use stratified train/test split and 5-fold cross-validation.
The data is imbalanced (20% churn) — suggest how to handle it and which metrics to report.
```

⚠ **Watch out** AI may suggest steps that cause **data leakage** (e.g. scaling before the train/test split) or invent library functions. Ask: *"Is there any data leakage risk in this code?"* and check the official docs.

10

Prompt Engineering for UI/UX

Research, ideas and words — you still own the design decisions.



Where AI helps designers

Research & personas

Draft interview questions, survey forms and starter personas to validate with real users.

Ideation & flows

Brainstorm features, user journeys and screen lists for wireframing.

UX writing

Button labels, error messages, onboarding text, empty states.

Accessibility & critique

WCAG checklists, colour-contrast guidance and heuristic reviews.

PROMPT · USER RESEARCH

```
I'm designing a mobile app that helps college students in Kerala find part-time jobs.
Write 10 open-ended user interview questions to understand their problems.
Avoid leading questions. Then suggest 3 assumptions I should test.
```

PROMPT · USER FLOW

```
Act as a senior UX designer. Map the user flow for a first-time user
signing up and applying to their first job in the app above.
List each screen, the main action on it, and possible drop-off points.
Format as a numbered list I can turn into a Figma wireframe.
```

PROMPT · MICROCOPY

```
Write 5 friendly, clear options for the error message shown when a
student uploads a resume larger than 5 MB. Max 12 words each.
Tone: helpful, not technical. Include what the user should do next.
```

PROMPT · DESIGN CRITIQUE

```
Here is a description of my login screen: [describe layout, colours, fonts].
Review it against Nielsen's 10 usability heuristics and WCAG 2.2 AA.
Give 5 specific, prioritised improvements.
```

💡 **Remember** AI-generated personas are **hypotheses, not research**. Talk to real users. Use AI to speed up preparation and writing — not to replace empathy and testing.

11 Golden Rules, Mistakes & Your 30-Day Plan

Small habits that make a big difference.

10 golden rules

1. Be specific — say exactly what you want
2. Give context — your level, project and tools
3. Mention versions — Python 3.12, React 19...
4. Ask for a format — table, steps, code + comments
5. Break big tasks into small prompts
6. Show examples of the output you want
7. Iterate — refine instead of restarting
8. Ask "why?" until you truly understand
9. Verify — run, test and check sources
10. Protect data — no secrets or personal info

Common mistakes students make

Mistake	Better approach
One-line vague prompts	Use the Role + Context + Task + Format formula
Trusting the first answer blindly	Test it, then ask "what could be wrong with this?"
Copy-pasting code into assignments	Use AI to learn; write and explain your own solution
Very long chats on many topics	Start a fresh chat for a new task to keep context clean
Not saying what went wrong	Paste the exact error / output and what you expected

Your 30-day practice plan

Week 1 · Foundations

Practise the 6-part formula daily. Rewrite 5 vague prompts into strong ones and compare results.

Week 2 · Your track

Pick your career section from this guide and use 3 of its prompts on a mini task every day.

Week 3 · Build

Build one small portfolio project with AI as pair programmer. Keep a "prompt diary" of what worked.

Week 4 · Showcase

Polish the project, write the README, and prepare to explain every line in an interview.

Reusable prompt template

COPY & FILL

```
Act as a [ROLE].
I am a [YOUR LEVEL] working on [PROJECT / CONTEXT] using [TOOLS + VERSIONS].
Your task: [CLEAR TASK].
Constraints: [RULES, LIMITS, WHAT TO AVOID].
Output format: [TABLE / CODE WITH COMMENTS / STEPS / WORD LIMIT].
Here is an example of what I want: [EXAMPLE].
Before you start, ask me any questions if something is unclear.
```

🎓 **Final word** Prompt engineering isn't about magic words. It's about **clear thinking** — knowing what you want, explaining it well, and checking the result. Master that, and every AI tool becomes a powerful partner in your IT career.